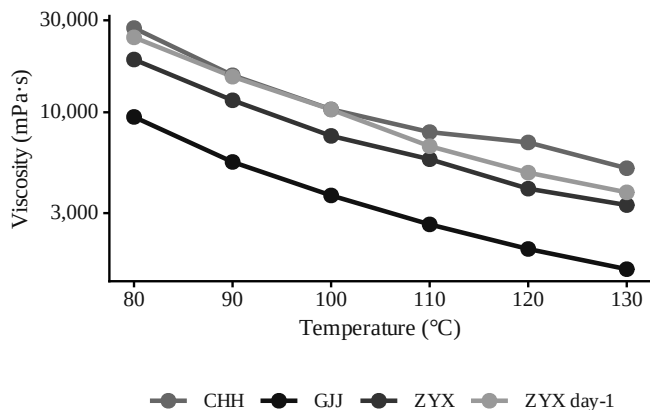


Figure 2 | Realization-dependent viscosity variation is dominated by a calibratable state shift

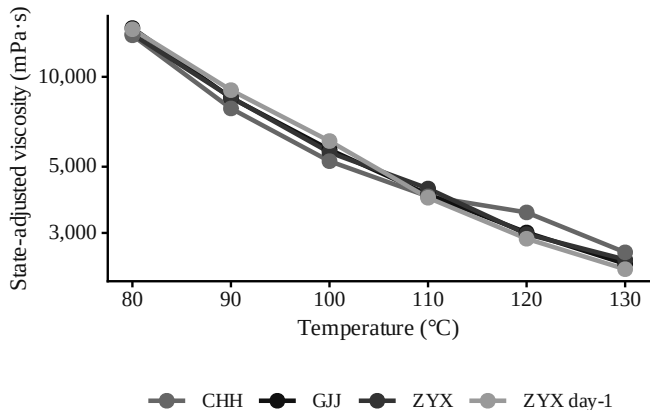
A Same recipe, different realized viscosity levels

Nominally identical E2 realizations remain separated across the full temperature sweep



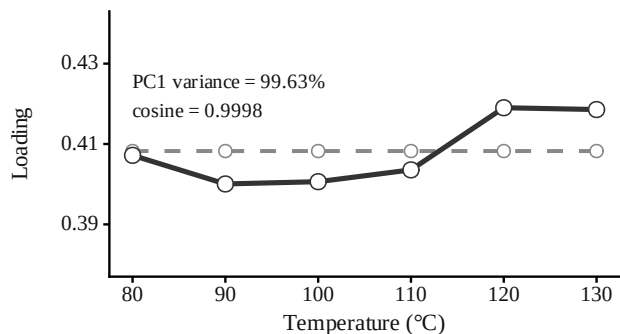
B State-offset correction collapses the curves

Realization-specific a_{fr} removed; shared thermal response retained



C The dominant state mode is a vertical shift

PC1 loading compared with an ideal constant displacement in log-viscosity space



D State conditioning restores predictive closure

Same quadratic thermal basis; leave-one-temperature-out error and

